



Follow-up Observations and an Improved Orbital Period for Exoplanet Candidate TOI-3505.01

Mason Cao¹, Annalyse Dickinson², Owen Alfaro², Rianne Eccleston², Kasey Davidson², Peter Plavchan²

¹Lambert High School, Suwanee, Georgia; ²Mason Space Center, Department of Physics and Astronomy, College of Science, George Mason University

Schar Scholars three month internship:
NASA Data Science Analysis and Scholarly Research Paper Writing

Abstract

TOI-3505.01 is a planet candidate found by NASA's TESS mission. It circles a 6220 K star that is 1.34 times the radius of the Sun (TIC 390988385, $V = 11.2$) every 2.9151 days. On July 22, 2022 we watched the star with the 0.8 m telescope at George Mason University during the hours the transit was supposed to happen. We took 283 images in R band and kept 263 good measurements, with a scatter of 2.50 ppt. The star never dimmed (-0.66 ± 0.39 ppt). To see whether that meant anything, we added fake transits to our data. A fake 2.91 ppt transit came back at 5.7σ , so we would have caught a real one.

We saw nothing because the prediction was off. With the period we have now, the closest transit was 17.5 hours before we started. So we went to the TESS archive and timed 27 transits in Sectors 14, 41, 54, and 81, a 5.05 year stretch. Fitting a line through those times gives $P = 2.9151516 \pm 0.0000049$ days. That is 2.4 times tighter than the TESS catalog period and 1.4 times tighter than the best published one, and it agrees with both. It also cuts the predicted transit time for 2026 from 9 minutes to 3.

Materials and Methods

Taking and measuring the images

- Mason 0.8 m telescope, July 22, 2022 UT. R filter, 50 second exposures, 0.362" per pixel. We got 283 images and 281 of them lined up into one 5.07 hour run.
- We subtracted darks, divided by flats, and shifted every image so the stars matched, using 25 reference stars, then redid each step in Python to check it. Astrometry.net matched our field to sky coordinates and landed 0.47 px from our measured position.
- We measured brightness in AstrolmageJ with a **25 px (9.1") aperture**, a **70–139 px sky ring**, and comparison stars C2–C11 picked from the Gaia DR3 list. We chose 25 px by trying every radius from 20 to 50 px and keeping the one with the least scatter. Our own Python photometry agrees with AstrolmageJ at $r = 0.990$.
- We cut 18 measurements for bad image quality and wrote down why for each, leaving **263**. No detrending passed the test we set up ahead of time, so the light curve is left as measured.

Working with the TESS archive

- We used QLP for Sectors 14 and 41, which is the only pipeline that covers them, and the SPOC 2-minute light curves for Sectors 54 and 81. Both pipelines measure the same photons, so we let only the better-sampled one speak for each sector.
- We ran a box least-squares search on each sector, then fit each transit one at a time to get 27 mid-transit times and fit a straight line through them. We fit a trapezoid rather than a flat box, because this transit is nearly grazing and spends most of its time in ingress and egress.
- We threw out one event in Sector 81 that sits on the edge of a data gap. It fills less than half the fit window, and with the middle missing the fit returns a 24 ppt dip where the real one is 3 ppt.

Checking the neighbours

- TFOP asks for every star inside 2.5', so we measured all 280 catalog stars in that circle bright enough to matter, on our own images, across the whole night.
- A star is cleared when our depth limit for it sits below the eclipse depth it would need to fake our signal. **222 of the 280** clear. Two sit inside our aperture, so we cannot separate them from the target at all.

Results

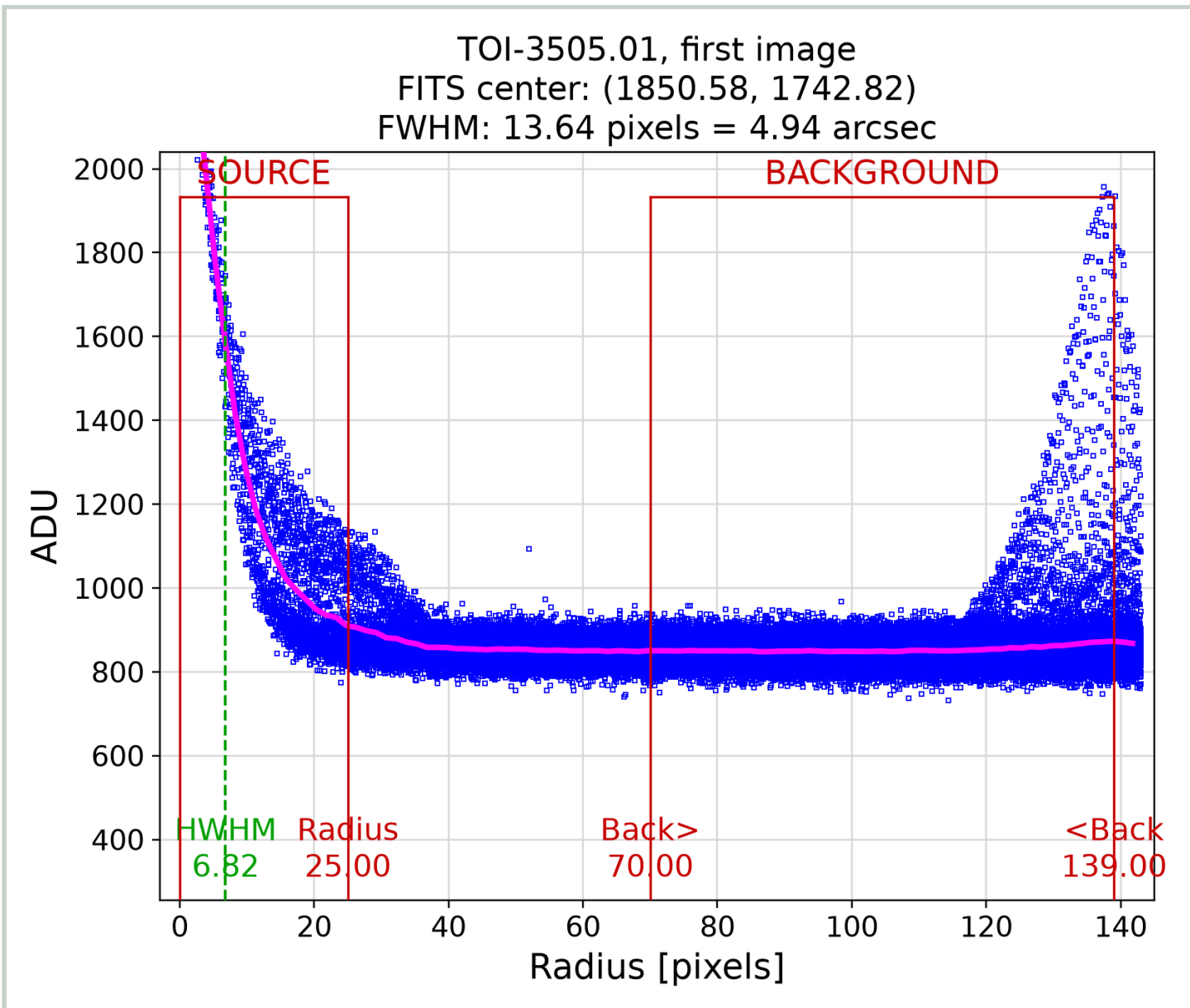


Fig. 1. Seeing profile of our star on the first image. The star's light flattens into the sky by about 25 px, which is where we set the aperture. FWHM is 13.6 px, or 4.94".

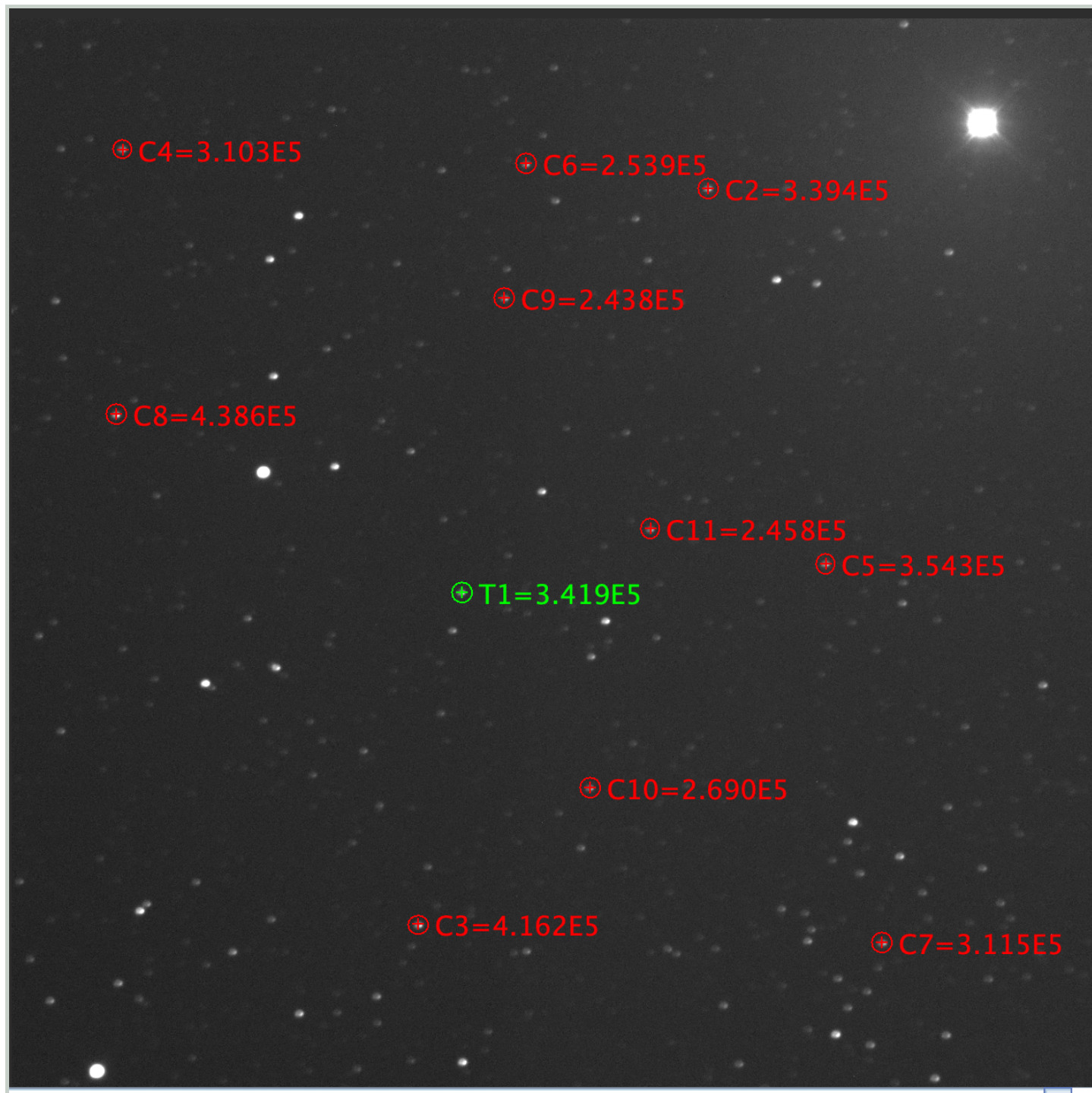


Fig. 2. Our field in AstrolmageJ. T1 is the target and C2–C11 are the comparison stars we picked off the Gaia DR3 list, with the counts each one gave.

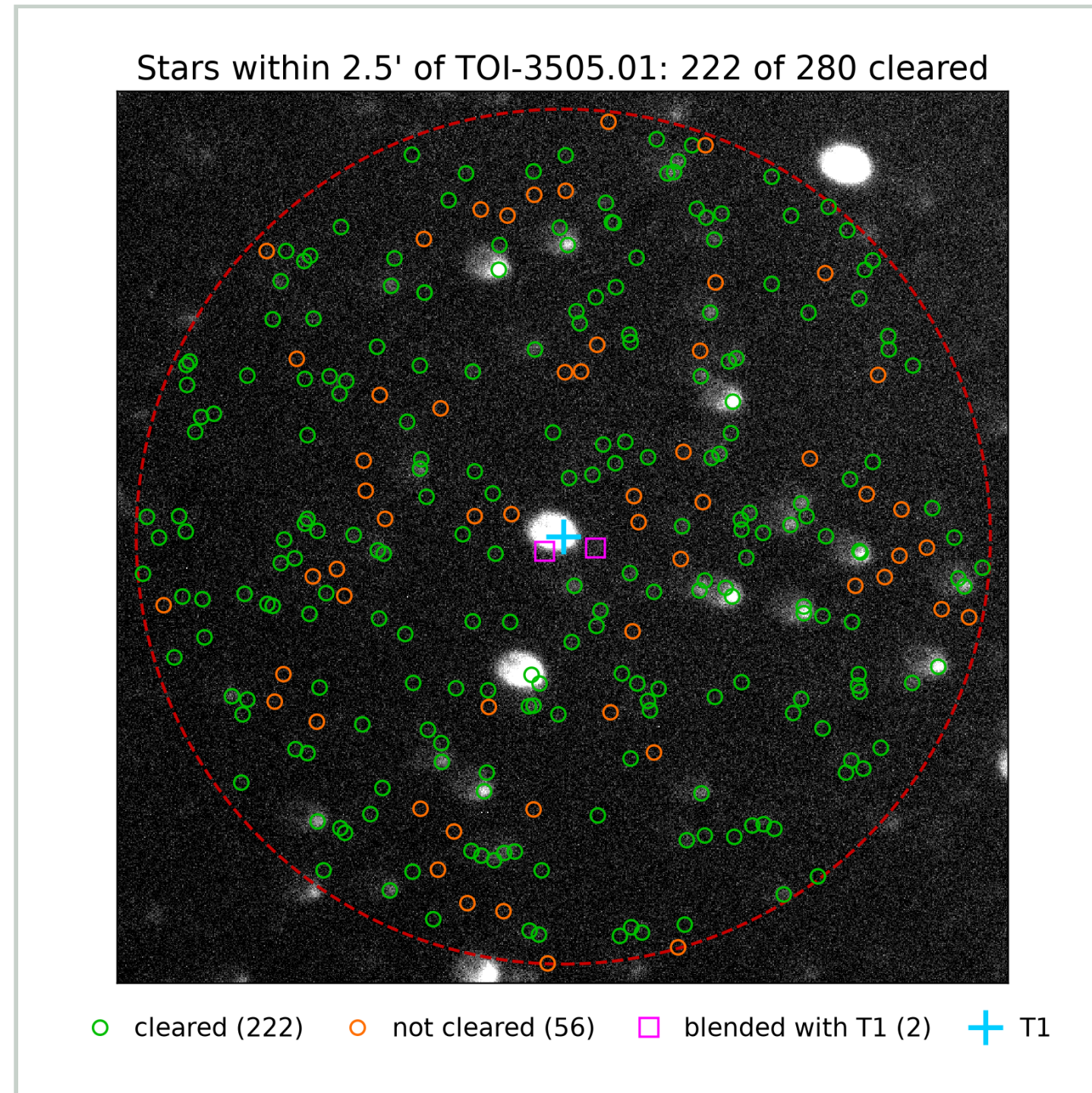


Fig. 3. Every catalog star inside 2.5' bright enough to fake our signal, measured on our own images. Green is cleared, orange is not. The two magenta squares sit inside our aperture, so we cannot separate them from the target.

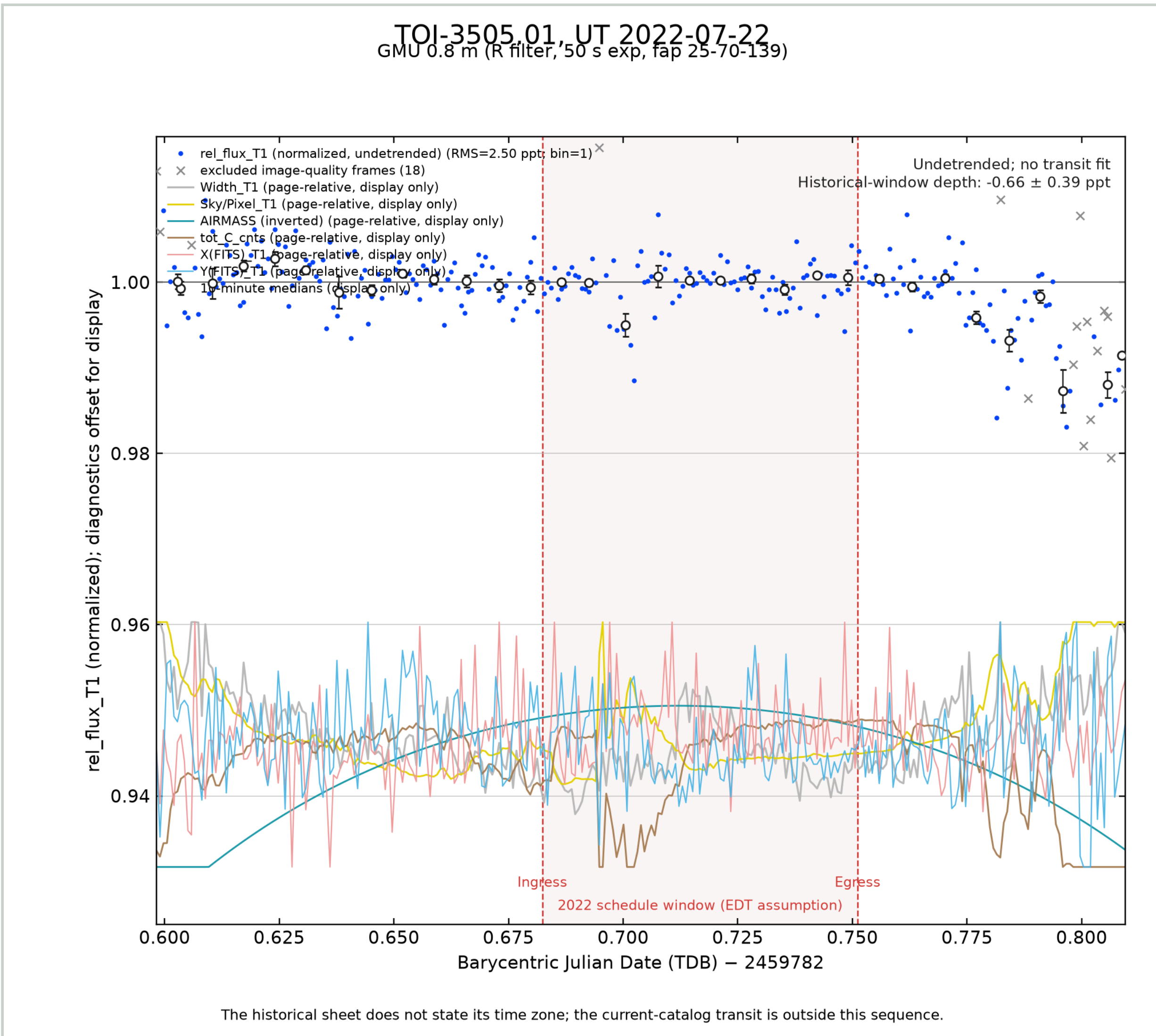


Fig. 4. Our light curve. 263 measurements at 2.50 ppt scatter. The shaded band is when the transit was scheduled and the red dashed lines are the predicted ingress and egress. Nothing dips there. The traces along the bottom are airmass, sky level, star width, and centroid position, which we checked for trends. None of them earned a detrend under our rules, so the curve is shown as measured.

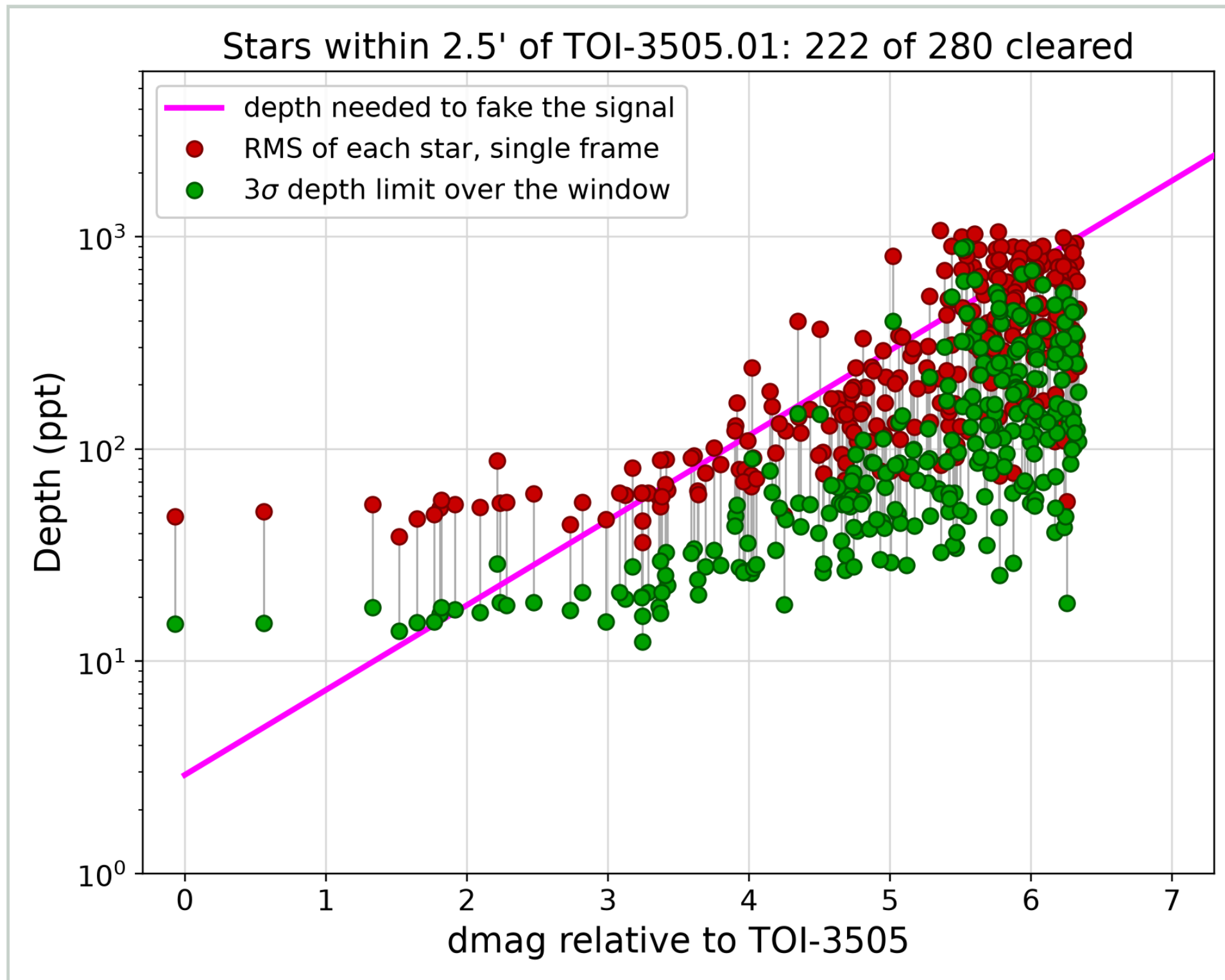


Fig. 5. Each neighbour's brightness difference against how deep we can see. The magenta line is the eclipse depth that star would need to fake our signal. Red is the scatter in a single frame, green is the depth limit after using the whole window, and the grey bars connect the same star. Green points below the line are cleared.

Discussion

Why we saw no transit

We picked this night using a 2022 prediction. With the period we have now, the nearest transit was 17.5 hours before we started, so there was nothing to catch. Our run also lands in a gap in TESS Sector 54, so we cannot line the two up. We added fake 2.91 ppt transits and pulled them back at 5.7σ , so the flat curve is a real limit, not noise.

What the TESS archive gave us

One night could not fix the prediction, so we used the archive instead. Timing 27 transits across four sectors gives $P = 2.9151516 \pm 0.0000049$ d. What buys the precision is the lever arm: our transits span 5.05 years, while the catalog row is built on three sectors and the official SPOC fit on two. So we land 2.4 times tighter than the catalog and 1.4 times tighter than SPOC, and we agree with both (0.3σ and 0.7σ). The scatter about the line is 8.5 min for $\chi^2/\text{dof} = 1.30$, so we widened the error bars by 1.14. Dropping Sector 14, where each image covers 30 minutes, moves the period by 0.6σ .

Why the planet's size is uncertain

In the official SPOC fit the planet barely clips the edge of the star ($b = 0.92$). For that to work the star would have to be 3.6 times less dense than a $1.34 R_{\odot}$, $1.25 M_{\odot}$ star can be. This close to the edge, depth, length, b , and $a/R_{\star}$ all trade off. Keeping the star's density and matching our 2.06 h length gives $b = 0.85$ and $a/R_{\star} = 6.93$. Either way the planet only grazes the star, so $9.0 R_{\oplus}$ is much less certain than its error bar looks.

What we cannot claim

Another star sits $0.517''$ away and adds about 21 percent of the primary's light. Adaptive optics on the Shane 3 m sees it at $\Delta J = 1.53$ and $\Delta Ks = 1.59$, so it is redder than our target. Neither our telescope nor TESS can split the pair, so if the transit is on that star the depth and radius are bigger than we quote. Our SPOC comparisons use the same TESS data we did, so they check our method rather than add a detection. We are not the first here either: TFOP holds six other ground light curves, imaging, and five spectra, none of it published. Nobody had fit an ephemeris across all four sectors.

Checks for False Alarms

Same depth in all four sectors	$\chi^2 = 4.3 / 3$
Odd and even transits match	$\leq 1.7\sigma$
No second dip half an orbit later	$S/N \leq 1.7$
Dimming centered on our star	$1.6\text{--}5.1''$
Compared with one TESS pixel	inside 21"
Nearby stars ruled out, 2.5'	222 of 280
SPOC odds of a false alarm	1×10^{-143}
Star $0.517''$ away	cannot separate

Conclusions and Future Work

- A better period.** Timing 27 transits over 5.05 years gives $P = 2.9151516 \pm 0.0000049$ d, tighter than both published values and agreeing with them, and the 2026 transit time is now good to 3 minutes instead of 9.
- Our night still counts.** The 2.50 ppt light curve rules out a transit during the window we watched, and our fake-transit tests show we would have seen one.
- Still a good candidate.** Nothing we checked points to a false alarm. The signal looks like a roughly Jupiter-sized planet grazing the edge of its star. What limits us now is that grazing path and the star $0.517''$ away, not our photometry.
- Next:** watch a transit on the new period, since a 3 minute window makes that worth doing on a 0.8 m telescope.
- Next:** fit a real limb-darkened model with MCMC and a density prior to pin down b and the planet's size.
- Next:** pull the four-colour photometry and the six Keck velocities other groups already filed and test them against this geometry. A depth that changes with colour would mean a blended eclipsing binary.

Acknowledgements

This research was made possible through the support of George Mason University's College of Science, which supports the Schar Scholars Program. We used data from NASA's TESS mission, which we got through the Mikulski Archive for Space Telescopes, and data from ESA's Gaia mission.

Major Citations

- Ricker, G. R. et al. 2015, *JATIS*, 1, 014003.
- Collins, K. A. et al. 2017, *AJ*, **153**, 77.
- Reefe, M. et al. 2022, *JATIS*, **8**, 027002.

- Huang, C. X. et al. 2020, *RNAAS*, **4**, 204.
- Jenkins, J. M. et al. 2016, *Proc. SPIE*, **9913**, 99133E.
- Twicken, J. D. et al. 2018, *PASP*, **130**, 064502.

- Mandel, K. & Agol, E. 2002, *ApJ*, **580**, L171.
- Stassun, K. G. et al. 2019, *AJ*, **158**, 138.